



DILEEP KUMAR THIRUVENKADAM

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SUMMARY

Software Engineering Master's student at Hochschule Heilbronn with a focus on DevSecOps, cloud infrastructure, and automation. With professional experience in backend development and test orchestration, I bridge the gap between development and operations. My academic performance includes a 1.0 grade in Cloud Computing and DevOps & SecOps. I am proficient in Python, Java, and JavaScript frameworks.

EDUCATION

Hochschule Heilbronn, Heilbronn, BW, Germany 09/2024-Present

- Current Average Grade: 2.2
- Notable Achievements: Grade 1.0 in Cloud Computing and DevOps & SecOps; Grade 1.3 in Advanced Software Architecture and Data Science Lab

SRM Institute of Science and Technology, Chennai, TN, Germany. 06/2018-05/2022

Final Grade: 1.9 (German Scale)

WORK EXPERIENCE

Quaking Aspen Private Limited, Associate Software Developer, Chennai, TN, India. 08/2022-07/2024

- Optimized core Python backend modules by refactoring query logic and implementing asynchronous processing, achieving a reduction in system latency.
- Architected comprehensive Python-based test automation suites, replacing manual testing workflows to reduce effort and ensure consistent release velocity across a 32-sprint, 96-week development cycle.
- Developed Python automation scripts to reduce manual tasks and improve operational efficiency. Technologies: Python, shell scripting, and cron jobs.
- Participated in sprint planning and retrospective within an Agile Scrum team; owned task lifecycle tracking in Jira and enforced Git branching standards to maintain codebase stability
- Contributed to the engineering team's quality standards by reviewing 500+ pull requests, enforcing idiomatic Python practices.

Zoho Corporation, Web Development Intern, Chennai, TN, India 01/2021-06/2021

- Designed and implemented responsive UI components using ReactJS, focusing on efficient state management and API data rendering
- Performed unit and integration testing to ensure cross-browser compatibility and UI consistency
- Delivered 12 production-ready UI features within a 5-member front-end team, meeting sprint deadlines across a 6 month engagement

PROJECTS & LEADERSHIP EXPERIENCE

COVID-19 Detection from Chest X-rays, Bachelor's Final year project 12/2021-05/2022

- Developed a Convolutional Neural Network (CNN) to classify lung X-ray images into COVID-positive, pneumonia, and normal cases, preprocessing datasets with OpenCV and image augmentation.
- Evaluated and visualized performance metrics including accuracy, precision, recall, F1 score, and confusion matrices using Matplotlib and Seaborn to ensure model reliability.

Automation for Workflow Optimization, Quacking Aspen, Chennai, TN 08/2022-02/2023

- Developed Python automation scripts to reduce manual tasks and improve operational efficiency. Technologies: Python, shell scripting, and cron jobs.

CI/CD Pipeline Integration, Hochschule Heilbronn (HHN) - Devops, Heilbronn, BW, Germany 09/2024-01/2025

- Integrated GitHub Actions for automated testing, deployment, and version control. Technologies: GitHub Actions, Docker, Terraform, AWS

Cloud Migration & Monitoring, Hochschule Heilbronn (HHN) - Devops, Heilbronn, BW, Germany 03/2025-07/2025

- Migrating services to AWS and setting up monitoring/logging tools. Technologies: AWS CloudWatch, Terraform, Docker

Intelligent Traffic Simulation System, HOCHSCHULE HEILBRONN - Realtime 3D Engine, Heilbronn, BW, Germany 09/2025-02/2025

- Developed a traffic flow modeling tool in Python to simulate real-world congestion scenarios and optimize urban routing algorithms.

SKILLS

Languages: English (Fluent), German (Intermediate), Tamil (Native)

Skills: Vulnerability Scanning, Testing, Terraform, SQL (PostgreSQL, MySQL), SonarQube (Static Analysis), Snowflake, Shift-Left Security, Selenium, Scikit-learn, RestAssured, Regression, Python, PyTest, Power BI, Pandas, NumPy, Manual UAT, Kubernetes, JUnit, Jira, JavaScript (Node.js, Express, ReactJS), Java (Spring Boot), Docker, Databricks, Cypress, CI/CD Pipeline Design, Bash scripting, and AWS (S3, CloudWatch, SNS, IAM, Elastic Beanstalk)